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A novel masking model for Buddhist literature understanding by using Generative Adversarial Networks

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ABSTRACT

This paper is focused on ancient Chinese Buddhist literature understanding. Buddhist literature incorporates a plethora of dialects and slang, which makes it challenging to extract semantic meaning seamlessly. To address this issue, a Generative Adversarial Network Masking Model (GAN-MM) is proposed to pre-train BERT models. This method focuses on optimizing the Masked Language Model (MLM) and takes advantage of the rich semantics of Buddhist terminologies as well as the absence of functional words in Buddhist literature. Furthermore, a semi-supervised learning algorithm was developed to train the GAN-MM. The model was evaluated by assessing its performance on two private tasks related to Buddhist literature understanding: sentiment classification and text segmentation, as well as two public tasks pertaining to the ancient Chinese understanding. The experimental results demonstrated a significant enhancement in the pre-training of BERT models when utilizing GAN-MM, as compared to conventional MLM methods. A large-scale Buddhist dataset, including 20,075 utterance documents and 146 million tokens, is public released at https://data.mendeley.com/datasets/5hzs8w46jh/1.

1. Introduction

The introduction of Bidirectional Encoder Representations from Transformers (BERT) by Devlin, Chang, Lee, and Toutanova (2018) revolutionized Natural Language Processing (NLP) tasks with remarkable performance. BERT utilized a *pre-training and fine-tuning* approach to tackle various NLP tasks. BERT inspired the development of several popular models, such as those proposed by Cui, Che, Liu, Qin, and Yang (2021), Diao, Bai, Song, Zhang, and Wang (2020), Lengkeek, van der Knaap, and Frasincar (2023), Sun, Huang, and Wei (2023), Sun et al. (2021), Suzuki, Sakaji, Hirano, and Izumi (2023), Zhang et al. (2021) and Zhu, Huang, Zhang, Zhu, and Huang (2020), designed to be pre-trained on large-scale Chinese corpora. Consequently, Chinese NLP tasks, such as machine reading comprehension (Cui et al., 2019), sentence classification (Conneau et al., 2018), text classification (Xu et al., 2020), and named entity recognition (Li et al., 2020), were performed using these models to achieve state-of-the-art results. BERT's pre-training process includes two objective functions: Masked Language Model (MLM) and Next Sentence Prediction (NSP). These tasks allow BERT to be pre-trained on unlabeled data, contributing to its success.

The ancient Chinese language is a valuable tool for studying the history and culture of ancient China, including historical records, ancient poems, Buddhist literature, and more (Zhuo & Zhang, 2024). However, investigating ancient Chinese can be challenging for those unfamiliar with the field of linguistics. Utilizing BERT-based methods to address NLP tasks related to ancient Chinese is practical and meaningful in unlocking the potential of ancient Chinese history and culture. Unfortunately, pre-training BERT models on modern Chinese corpora

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does not yield satisfactory results for ancient Chinese due to the language differences between modern and ancient Chinese. While modern Chinese is straightforward, ancient Chinese is complex and abstruse, with component words that have diverse and intricate semantics.

Pre-training BERT models on ancient Chinese corpora has emerged as a novel approach in recent years (Hu, Sun, Liu, Liu, & Wang, 2024; Li, 2023; Sommerschield et al., 2023; Xiang, Liu, Li, Qiu, & Hu, 2024; Yang et al., 2024; Zhu et al., 2023; Zhuo & Zhang, 2024). Tian, Yang, Liu, and Lv (2021) built a corpus from ancient poems and couplets to pre-train an Ancient Chinese BERT (AnchiBERT) model for language understanding and generation. Similarly, Yang, Chen, and Chen (2021) built a corpus of ancient Chinese books to pre-train the BERT model, named GuwenUNILM, for ancient–modern Chinese translation. Siku-BERT (Wang et al., 2022) is another released Chinese BERT model pre-trained on Siku Quanshu full-text corpora, which contains about 536 million Chinese Characters. The model achieved good performance on downstream tasks of automatic word segmentation, segmentation punctuation, part-of-speech tagging, and named entity recognition for Zuo Zhuan corpora. BERT-ancient (Wang, 2022) was pre-trained on six times larger corpora than the Siku Quanshu corpora. The used vocabulary size was expanded to 38,208. BERT-ancient achieved noteworthy downstream performance for Chinese word segmentation and part-of-speech tagging on the competition EvaHan 2022.[1]

This paper is focused on addressing challenges related to NLP tasks when analyzing ancient Chinese Buddhist literature. To leverage the acquired knowledge of functional words and Buddhist terminologies, we propose a novel model, the Generative Adversarial Network Masking Model (GAN-MM), as an alternative to the MLM for pre-training BERT. GAN-MM can selectively mask crucial notional words by utilizing a semi-supervised learning algorithm. To the best of our knowledge, this paper is the first work to retrain BERT models in Buddhist literature. The key contributions of this paper are summarized below:

1. The GAN-MM is an innovative masking method for pre-training BERT, where the generator's objective is to recover the embedding features from the masked fake sample, while the discriminator's objective is to assess the difficulty of the recovery. GAN-MM significantly speeds up the pre-training process and helps identify samples that mask crucial notional words.
2. A semi-supervised learning algorithm has been proposed to train the GAN-MM. It is a balance between masking Buddhist terminologies and functional words by assigning a combined penalization factor to the reconstruction loss for training the generator.
3. A corpus of Buddhist literature was collected from the CBETA database,[2] including 23 utterance categories, 20,075 utterance documents, and 146 million tokens. The GAN-MM performed well on two private Buddhist understanding tasks and two public ancient Chinese NLP tasks, outperforming general MLM methods. The dataset is available online[3] and released by He (2024).

In the rest of this paper, we first introduce the ancient Chinese Buddhist language understanding problem in Section 2. Then, we introduce the domain-specific BERT-based models, variants of MLM, and GAN in Section 3. The proposed method is explained in Section 4. The datasets are introduced in Section 5. The experimental results and analysis are introduced in Section 6. Finally, we conclude and discuss the proposed method in Section 7.

2. Problem formulation

Buddhist literature is a hybrid cultural product that emerged from the synthesis of foreign Buddhist culture within the ancient Chinese context. It represents an essential component of the Chinese cultural heritage, particularly during the Tang and Song Dynasties, with significant implications for modern Chinese heuristic studies.

Buddhist literature is more difficult to understand than general ancient Chinese language for two primary reasons. First, Buddhist literature includes a significant amount of colloquial and informal expressions. This style reflects its unofficial nature and is characterized by the presence of dialects and slang. In contrast, general ancient Chinese forms such as verse, couplets, and odes typically employ a stylized language format, making them easier for large language models to comprehend. Second, accurately interpreting Buddhist literature requires a deep understanding of its semantic context, as many texts contain implied meanings that go beyond the literal text and demand nuanced interpretation. Overall, this paper aims to provide insight into the unique challenges of understanding ancient Buddhist literature, emphasizing the need for advanced semantic context knowledge and novel techniques to address these challenges.

Ancient Buddhist literature understanding with pre-training BERT models is challenging due to the semantic complexity of the ancient Chinese language. In Ancient Chinese, words can be categorized as notional or functional based on their role within a sentence. Functional words, which include conjunctions, auxiliary words, and modal particles, lack clear semantics and are primarily used to support the grammatical structure of a sentence. These words are often positioned at the beginning or end of a sentence. Notional words typically carry meaningful semantics within a sentence. In the context of Buddhist literature, some key notional words hold significant Buddhist meanings, often referred to as Buddhist terminologies. Buddhist terminologies are rich in semantic content and play a crucial role in understanding the philosophical nuances present within the text. The different semantic representations between notional and functional words[4] motivate our proposed method in this paper.

3. Related work

3.1. Domain-specific BERT models

Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2018) was designed to be pre-trained on large-scale corpora via unsupervised learning. The pre-training stage consists of two tasks: Masked Language Model (MLM) and Next Sentence Prediction (NSP). MLM randomly masks some of the word tokens in a sentence and aims to predict the source word tokens on the context features. NSP aims to predict whether one sentence is followed by another sentence. Both tasks require no more labeling work for network training, so BERT can be pre-trained on very large-scale datasets. It has been evaluated that BERT has robust performance in a wide range of downstream NLP tasks.

Recently, BERT models have been proposed to cope with some domain-specific tasks. To extract valuable knowledge from biomedical documents, a biomedical language model, named BioBERT (Lee et al., 2020), was proposed to be pre-trained on large-scale biomedical corpora and outperformed baseline models on three downstream biomedical text mining tasks. Alsentzer et al. (2019) optimized basic BERT and BioBERT by pre-training them on clinical text of all note types and discharge summaries, which verified that the domain-specific contextual embeddings improved experimental results. In addition, the BERT models are mostly proposed to cope with specific monolingual language tasks besides English. Martin et al. (2020) proposed the

[1] https://circse.github.io/LT4HALA/2022/EvaHan.
[2] http://www.cbeta.org/.
[3] https://data.mendeley.com/datasets/5hzs8w46jh/1.
[4] https://github.com/ithet1007/GAN-MM-examples/blob/master/Fig1.png.

CamemBERT to be trained on French corpora, which was crawled from Wikipedia. CamemBERT reached wonderful performance on natural language inference, dependency parsing, named entity recognition, and part-of-speech tagging tasks. Delobelle, Winters, and Berendt (2020) proposed the RobBERT to train a Dutch language model and evaluated its effectiveness on small domain-specific datasets. AraBERT (Antoun, Baly, & Hajj, 2020) was released to cope with Arabic sentiment analysis, question answering, and named entity recognition.

As Chinese is a widely used language, Chinese BERT models have been well exploited. To learn the potential word or phrase boundary information in Chinese, Diao et al. (2020) proposed a model to incorporate the information of words and phrases, which enhanced the BERT-based text encoder by N-gram representations. Sun et al. (2021) deeply analyzed the properties of Chinese and proposed ChineseBERT by incorporating the information of pinyin and glyph, where the glyph information captured the semantic visual features and the pinyin information used the features of heteronym phenomenon in Chinese. Cui et al. (2021) introduced the whole word masking strategy to pre-train BERT-based models for Chinese, which involved the semantics of the whole word. In the ancient Chinese domain, AnchiBERT (Tian et al., 2021) focused on the regular syntactic form of ancient poems and couplets, and was fine-tuned for downstream language understanding and generation. Yang et al. (2021) built a GuwenUNILM model to translate ancient Chinese literature to modern Chinese and was trained on an open sentence-level parallel corpus. GuwenBERT[5] was a pre-trained BERT model for named entity recognition of ancient Chinese on the corpora of ancient books.

3.2. Variants of masked language model

The success of BERT is due to its pre-training tasks of MLM and NSP. ALBERT (Lan et al., 2019) proposed to replace traditional NSP tasks with sentence-order prediction tasks. By carefully studying the function of various components in BERT, Liu et al. (2019) concluded that masking strategy is the key factor for pre-training BERT and suggested removing the NSP task. They proved that training longer with longer sentences over more data is of great importance. MLM aimed to predict masked tokens using the bidirectional context of sentences rather than left-to-right or right-to-left context. 15% of all tokens were masked randomly and then were predicted. In fact, the situation of masked tokens would not happened during fine-tuning. To alleviate the mismatch between pre-training and fine-tuning, the masked tokens were replaced with [MASK] tokens 80% of the time, random tokens 10% of the time, and the source tokens 10% of the time.

To further resolve the above mismatch issue, Dai et al. (2019) proposed the Transformer-XL, which was pre-trained on unfixed-length context by an attentive language model. The segment-level recurrence mechanism and positional encoding scheme were used to model very long-term dependency. Inspired by Transformer-XL, Yang et al. (2019) proposed a generalized autoregressive pre-training method to overcome the mismatch problem by maximizing the expected likelihood over all permutations of the factorization order.

In recent years, the MLM become the highlight to be improved for pre-training BERT. As far as we know, BERT can segment a word into wordpiece tokens, which makes the prediction task easier. In 2019, Devlin et al. (2018) released the updated version of BERT by using the Whole Word Masking (WWM) strategy, which used the whole word as a masking unit. Cui et al. (2021) evaluated WWM for a series of Chinese BERT. Sun et al. (2019) enhanced the masking strategy by entity-level masking and phrase-level masking. The entity-level strategy masked the entities and phrase-level strategy masked the whole phrase which was generally used as a conceptual unit. MacBERT (Cui et al., 2020) proposed to use similar words to mask selected whole word tokens. Specifically, 15% words were selected for masking, where 80% of them were replaced with similar words, 10% of them were replaced with random words, and the rest kept the original words.

3.3. Generative adversarial networks for text generation

GAN was first proposed for image generation by Goodfellow et al. (2020). More recently, there has been a significant surge of interest in employing GANs for text generation. One approach in this domain involves utilizing recurrent neural networks (RNNs) as the foundational network structure for GANs. Zhang, Gan, and Carin (2016) employed a Long Short-Term Memory (LSTM) to extract temporal features and constructed a GAN for adversarial training to generate realistic text. Liang et al. (2021) proposed a text GAN framework for creative essay recommendation. The network was constructed by a BiLSTM and trained in an unsupervised end-to-end manner. Li, Pan, Wang, Yang, and Cambria (2018) explored GANs for category-based text generation. Their approach involved utilizing LSTM to build the network architecture and formulating the loss function based on prior category information, which led to improved generalization accuracy.

GAN were also jointly trained with other learning methods, such as self-attention, reinforcement learning, and loss truncation. Xu et al. (2018) introduced an advanced attentional GAN, which used an attention-driven multi-stage refinement algorithm for fine-gained text-to-image generation. Li et al. (2017) transformed text generation to a reinforcement learning problem and solved it by a policy gradient training with a GAN, where the reward was associated with every generation step. Kang and Hashimoto (2020) proposed a loss truncation method for training GAN on distinguishability on a summarization task.

4. Method

4.1. Motivation

The motivation behind this paper stems from a fundamental truth: the semantics of Ancient Chinese are significantly influenced by the masking of notional words rather than functional words.[6] Consequently, we endeavor to enhance the MLM during the pre-training of a language model based on these observations. Frequently masking notional words, especially Buddhist terminologies, holds the potential to benefit the pre-training of a language model. However, two key challenges must be taken into account. First, it is essential to determine the optimal frequency of masking notional words. Striking the right balance is crucial, as excessive masking of notional words can hinder the training of the language model. On the other hand, functional words, although only loosely related to the semantics of Ancient Chinese, still play a vital role in understanding language grammar and sentence structures. Therefore, some functional words should still be masked. Second, distinguishing between notional and functional words is not an easy manual task due to the inherent ambiguity of Ancient Chinese vocabulary. Labeling all notional and functional words in Buddhist literature would be a time-consuming and impractical endeavor.

In this paper, we endeavor to enhance MLM by addressing the aforementioned two challenges. To tackle the first challenge, we introduce an advanced Generative Adversarial Network Masking Model (GAN-MM). The GAN-MM process comprises two steps: firstly, oversampling m masked samples from a source sentence using the MLM strategy, and secondly, employing a trained GAN to select the most suitable sample based on the level of difficulty in reconstruction. To address the second challenge, we propose a semi-supervised learning algorithm aimed at balancing the masking frequency between notional and functional words during the training of GAN-MM. Detailed information regarding the proposed method is provided in the subsequent subsections.

[5] https://github.com/ethan-yt/guwenbert.

[6] https://github.com/ithet1007/GAN-MM-examples/blob/master/Fig2.png.

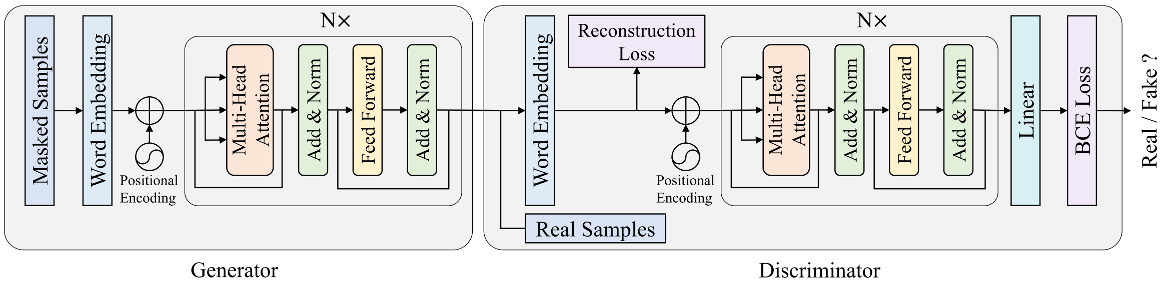


Fig. 1. The proposed GAN-MM network structure. The generator consists of a word embedding layer, a positional encoding layer, and N Transformer blocks. The discriminator has the same transformer blocks as the generator but followed by a linear layer.

4.2. Generative adversarial network masking model

The proposed GAN-MM is depicted in Fig. 1. The generator comprises a word embedding layer, a positional encoding layer, and N Transformer blocks. Its objective is to reconstruct a sentence from a masked sample. The masked samples are treated as fake samples and are input into the generator, producing predicted word embeddings. The generator then reconstructs the word embeddings from pairs of real and masked samples by minimizing the reconstruction loss. The discriminator employs the same Transformer blocks as the generator but is followed by a linear layer. Its role is to discriminate whether a sample is from the real world or oversampled by masking.

During the generator training stage, the real sample undergoes masking to serve as the input, and a reconstruction loss is formulated to penalize disparities between the original embedding and the generated word embedding of the real sample. Simultaneously, a Binary Cross Entropy (BCE) loss is established to deceive the discriminator, thereby ensuring that the generated results exhibit comparable statistical characteristics to real data. In the discriminator training stage, a set of generated n fake samples along with their corresponding real samples are supplied to the discriminator, aiming to enhance the discriminator's discrimination capabilities.

The proposed GAN-MM offers two innovative advantages in the selection of the best-masked token. Firstly, the generator's ability to recover indicates the level of semantic information missing from an Ancient Chinese sentence. When a masked token cannot be effectively recovered, the discriminator assigns a low score, designating this masked sample as a suitable candidate for feeding into a language model. Secondly, to fully harness the varying masking frequencies between notional and functional words, we can dynamically choose the real samples for training the generator. We suggest augmenting the reconstruction loss using a semi-supervised learning algorithm that takes into account the masking of notional and functional words. The specifics of this proposed algorithm are presented in the subsequent subsection.

4.3. Semi-supervised learning algorithm

The naive approach to training the generator involves assigning a substantial penalization factor to the reconstruction loss when a notional word is masked while applying a smaller penalization factor on the contrary. However, distinguishing between notional words and functional words is a challenging task due to the inherent ambiguity of Ancient Chinese. In this paper, we introduce a semi-supervised learning algorithm for the penalization of the reconstruction loss. We manually construct two dictionaries: one for commonly used Buddhist terminology, denoted as D_z, and another for commonly used ancient functional words, denoted as D_f. It is noteworthy that punctuation tokens are categorized as functional words within the functional word dictionary. Our approach begins by generating a set of m masked samples based on the Masked Language Model (MLM) rule, where approximately 15% of the tokens are masked. Among these, around 80% are replaced with the [MASK] token, 10% are replaced with random tokens, and the remaining 10% remain unchanged. After processing these m masked samples, we design a penalization score strategy for each sample i, where $1 \le i \le m$. Suppose the ith sample has K masked tokens, the penalization score strategy is as follows:

$$s_k = \begin{cases} 0.1, \text{ if } k\text{th word} \in D_f; \\ 0.5, \text{ if } k\text{th word} \notin D_f \cup D_z; \\ 0.9. \text{ if } k\text{th word} \in D_z. \end{cases} \tag{1}$$

where $1 \le k \le K$. The penalization factor can be summarized as follows:

$$\alpha = \frac{1}{K}\sum_{k=1}^{K} s_k. \tag{2}$$

Let the generator and the discriminator be G and D, respectively. Given a data pair $\{\mathbf{x}, \mathbf{z}\}$, where $\mathbf{x}$ indicates the real sample and $\mathbf{z}$ indicates the masked sample, respectively. Define the embedding layer as E, the prediction of the generator is $G\left(E(\mathbf{z};\theta_e);\theta_g\right)$, where θ_e and θ_g indicates the parameters of the embedding layer and G, respectively. To train the generator, we use the L1 loss to recover the masked sample as follows:

$$L_{rec} = \frac{1}{n}\sum_{i=1}^{n} \left| G\left(E(\mathbf{z}_i)\right) - E(\mathbf{x}_i) \right|, \tag{3}$$

where n is the minibatch size. The generator aims to fool the discriminator as follows:

$$L_g = \frac{1}{n}\sum_{i=1}^{n} log\left(1 - D\left(G(E(\mathbf{z}_i))\right)\right). \tag{4}$$

The total training loss for the generator is

$$\hat{L}_g = \alpha \cdot \beta \cdot L_{rec} + L_g, \tag{5}$$

where α is the penalization factor defined in Eq. (2) and β is a learning factor, respectively. We empirically set $\beta = 0.005$ in our experiments. It is worth noting that the generator and the discriminator share the embedding layer to obtain the embedding feature. To train the discriminator, the parameters of the generator are frozen and update the discriminator by

$$L_d = \frac{1}{n}\sum_{i=1}^{n} \left[log D\left(E(\mathbf{x}_i)\right) + log\left(1 - D\left(G(E(\mathbf{z}_i))\right)\right)\right]. \tag{6}$$

After the GAN-MM is trained, a masked sample of a minibatch is selected by Algorithm 1. It is worth noting two phenomena observed during network training. First, if a masked sample is generated with an excessively high frequency of masking notional words, the GAN-MM struggles to reconstruct the source sample, and this masked sample will not be selected as the final one according to Algorithm 1. Second, if a masked sample is generated with an excessively low frequency of masking notional words, the GAN-MM easily reconstructs the source sample, but this masked sample does not contribute to the network training. Therefore, after the GAN-MM is trained, a candidate masked sample with an optimal frequency of masking notional words is selected as the final one to pre-train the BERT. The experimental results, illustrated in Fig. 4, confirm our explanation.

Algorithm 1: GAN-MM sample selection.

Input : Trained GAN-MM G and D, candidate masked sample set C, and number of candidate masked samples m.
Output: Selected sample r.

```
y_min = 1;
for i = 1, 2, ..., m do
    y_i = D(G(c_i)), c_i ∈ C;
    if y_i < y_min then
        y_min = y_i;
        r = c_i;
    end
end
return r.
```

5. Datasets

Buddhist dataset. The original corpus was extracted from CBETA,[7] comprising 23 categories, 20,075 documents, and 146 million words, as detailed in Table 1. Following data preprocessing, which involved removing empty documents and non-Chinese characters, we transformed ancient Chinese characters into simplified characters using OpenCC.[8] We utilized the majority of this corpus to form the Buddhist pre-training dataset, with the exception of 3001 Zen documents, which were reserved for creating the downstream automatic Zen Text Segmentation (ZTS) and Zen Sentiment Classification (ZSC) datasets. In the end, the pre-training dataset comprises 10,119 documents. These documents were splited into text fragments with the maximum token length of 512. We do not apply any tokenization method and each Chinese character was represented as a token based on the granularity of ancient Chinese character, following Tian et al. (2021). 80% of these texts were allocated for model pre-training, while the remaining portion was used for model validation.

Buddhist dictionary. Functional words are among the most frequently utilized words in the realm of Chinese characters. The compilation of functional words is indispensable for comprehending and interpreting Ancient Chinese text, serving as a vital tool for the study of Ancient Chinese grammar and literature. However, the utilization of functional words is highly adaptable, and many functional words found in Ancient Chinese have fallen into disuse in modern Chinese, posing challenges to their compilation. Fortunately, Jie, Cui, and Zheng (2008) has meticulously organized influential works on functional words from the past, resulting in a collection of 1064 words that form the basis of the functional word dictionary utilized in this paper. In the case of collecting Buddhist terminologies, we referred to Ding (2011), which encompassed a wide array of Buddhist terminology, allusions, references to renowned monks, historical sites, and more. This resource not only identifies the part of speech for each term but also provides comprehensive explanations of their meanings, citing their sources. From this, we have extracted a set of 3187 widely used words, forming our Buddhist terminology dictionary.

Zen Sentiment Classification (ZSC) dataset. Zen literature is replete with numerous allusions and instances of verbal irony. Consequently, the sentiment classification of Zen texts presents a greater challenge compared to modern Chinese texts. Building on the prior research (Wang & Lei, 2010), we gathered 2795 negative texts and 2141 positive texts from Zen literature. Of these, 3436 texts were allocated for training, 500 for validation, and 1000 for testing purposes, respectively.

Table 1
The statistics of Buddhist utterances.

Utterances	No. of Docs	Utterances	No. of Docs
Agama Sutra	400	Madhyamaka	132
Apadana	340	Yoga	602
Prajnaparamita-sutra	1215	Analecta	346
Lotus Sutra	877	Pure Land Buddhism	348
Avatamsaka	983	Zen ministry	3001
Ratnakuta	331	History and biography	1509
Nirvana	357	Affairs	836
Mahasanghika	200	Dunhuang manuscripts	219
Vimalakirta	1232	Rare book	248
Tantric Theravada	1385	Tipitaka	331
Samaya	945	New editorial	3443
Bitanbe	795	Total	20,075

Zen Text Segmentation (ZTS) dataset. From the original pool of 3001 Zen documents, we crafted an automatic Zen Text Segmentation (ZTS) dataset. We excluded exceedingly short documents and those lacking punctuation. Each document was divided into multiple texts, each with a maximum length of 512 tokens. These texts were segmented based on the presence of 15 punctuation marks. In the end, the ZTS dataset comprises 28,862 training texts, 3966 validation texts, and 8501 testing texts.

Poem Topic Classification (PTC) dataset. The task of ancient Chinese Poem Topic Classification (PTC) is designed to classify poems into their respective literary topics. The dataset[9] comprises 3.2K classical Chinese poems, with 2.8K for training, 0.2K for validation, and 0.2K for testing, respectively. Each poem consists of four lines and is accompanied by titles and keywords. Furthermore, each poem is annotated with one of 9 literary topics.

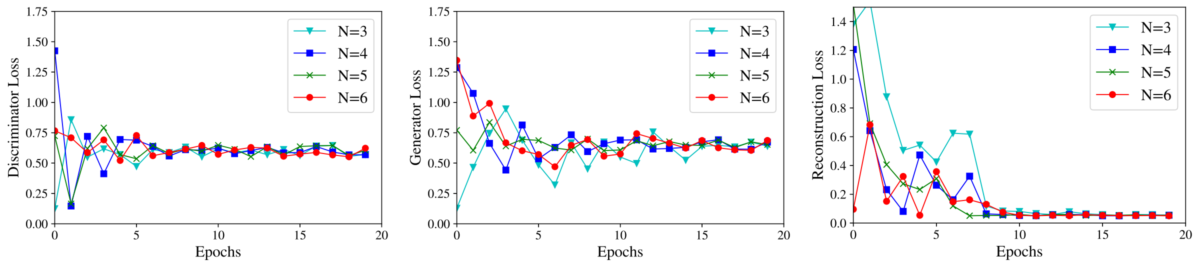
Chinese Couplet Generation (CCG) dataset. CCG is a task focused on generating the second sentence, often referred to as the subsequent clause, of a couplet based on the first sentence. The dataset[10] encompasses 0.77 million couplet pairs designated for training, along with 4000 couplet pairs for validation and an additional 4000 couplet pairs for testing.

6. Experiments

All experiments in this paper were implemented using PyTorch version 1.13 and were executed on four NVIDIA 3090 GPUs. The Hugging Face Transformers[11] framework was employed as the model toolkit and provides support for the original pre-trained models. In all experiments, a maximum sentence length of 512 tokens was utilized. We conducted a comparative analysis involving three baseline BERT models, namely BERT-base (Devlin et al., 2018), MacBERT-base (Cui et al., 2020), and SIKU-BERT (Wang et al., 2022). Additionally, three pre-trained models were included: BERT-wwm (Cui et al., 2021), BERT-ancient (Wang, 2022), and AnchiBERT (Tian et al., 2021), pre-trained using the Masked Language Model (MLM) and the proposed Generative Adversarial Network Masking Model (GAN-MM). For training GAN-MM, a batch size of 64 was employed, while for all other experiments, a batch size of 32 was used.

During training the GAN-MM, we used the univariate sensitivity analysis method to evaluate two important hyperparameters: the number of the Transformer blocks N and the candidate samples m. We used the Random Search method within 5 epochs to determine the learning rate and the weight decay. During the pre-training and downstream transfer learning stages, we directly follow the settings of compared SOTA methods, such as Anchi-BERT, BERT-wwm, and BERT-ancient.

[7] http://www.cbeta.org/.
[8] https://github.com/BYVoid/OpenCC.
[9] https://github.com/shuizhonghaitong/classification_GAT/tree/master/data.
[10] https://github.com/wb14123/couplet-dataset.
[11] https://huggingface.co/docs/transformers/index.



(a) The curves of the discriminator loss. (b) The curves of the generator loss. (c) The curves of the reconstruction loss.

Fig. 2. The curves of the validation loss when training GAN-MM.

	Masked candidate samples	Conf. $n_f:n_z$
1	有时水大与□大相触，有时水大□风大相触。以是缘故，虚空云中有雷声起。相师占□有五因缘不可□知，使占者迷惑。云何为都？□者云有雷电，占□当雨，以火大多故，烧云不□，□为占师□迷惑缘。二□云有雷第，占□当雨，有大风起，吹□四散，入诸山□，以□缘故，相□迷惑。	57.32% 1:14 ☑
2	有□水大割火大相□，有时水大与风大相□。以是缘故，虚空云中□雷声起。相师□雨有五□缘□可定知，使占者迷惑。云□为五？一者云有雷电，占□当雨□以火大法故，烧云不□，是为占师初□惑缘。二者□有雷电□占谓当雨，有大风起，吹云四散，□诸山间，以此缘故，相师迷惑。	60.05% 4:11
3	有时水□与火大相触，有时水□与风大相触。以是□故□虚空云中有□声起。相师占□有五因缘不可定知，□占者迷撒。云何□五？一□云有雷电，占谓□雨，□火大多故，烧云不雨，□为占师初迷惑缘。二□云有雷电，占□当雨，有大风起，吹云四散，□诸山间，以此缘的，相师迷惑。	59.34% 3:12
4	有时水大与火大相触，有时若大与风大相触。以是缘故，虚空□中有雷声起。□师占雨□五因缘不□定知，使□者迷惑□云何为五□一者□有雷电，占谓当雨，以□□多故，烧云不雨，是为占师初迷□缘。二者云有雷电，占谓当雨，有大风起，□云四散□入诸山□，□此缘故，相师迷惑非	63.92% 4:11
5	有时□大与火□相触，有时水大□风大相□。以□缘激，虚空云中有雷□起。相师占雨有五因缘不可定知□使占者迷惑。云何为五？一者云有篇电，占谓□雨，□火大多故，烧云不□，是为占□初迷惑缘。□者云有雷电□占谓当雨，有大风起，吹□四散，入□山间，以此缘故，相师迷惑。	68.45% 6:9

(a) Example 1.

	Masked candidate samples	Conf. $n_f:n_z$
1	疏心境者，□心之所发，由□之所牵。虽分□法体唯本觉无二别也。□名体相。有力者□妄心□故，本觉势强□力猛盛也。昔以翻流，则妄有力而真劣，□既反□□则真熏功盛而妄心势衰也。论自信己身□者，约人所跑故言己身，以己真如□自妄心有势□故，遂能□照。	62.83% 5:9
2	□心境者，由心之□发，由境之所牵。虽分二法体唯本觉无二别也。亦名体相。有力者，妄心劣故□本觉势强熏力猛盛也□昔卡随流，则妄有□而真劣，□既反流，则真□功盛而妄心势衰□。论自信□身等者，约□所说故言范身，以己真□熏自□心有势力□，遂□反照。	59.01% 4:10 ☑
3	疏心境者，阀心之所发，由境之所牵□虽分二法体唯本觉无二□也。亦名体相。□力者，妄心劣故，□觉势强□力猛盛也。昔□随流，则妄有力而真□，今□反流，则□熏功盛而妄□势衰也。论自信己身等□，约人所菲故□己身，以己真如熏自妄心有势□故，遂能□照。	59.32% 4:10
4	疏心境□，由心之□发，由境之所牵□虽分二法体唯本觉无二别也。亦名体相将有力者，妄心劣故，□觉势强熏力猛盛也。昔以□流，则妄有□而□劣□今既反流，则真熏功□而□□势衰也。论自信己身□者，约人飒□故言己身，以己真如熏自妄□有势力故，遂能反照。	61.59% 5:9
5	疏心境者，由心之所□，由境之□牵。虽分□法体唯本觉无二别也。亦名体相□有力者，妄心达故，本觉势强熏力猛盛也。昔以□流□则妄有力□真劣，今既□流，则□熏功□□妄心势衰□。论自信己身等者，约人所说故言己身，以己真如熏自□心有势力□，遂能反复。	67.45% 7:7

(b) Example 2.

Fig. 3. Two examples of selecting final samples from the masked candidate samples, where $m = 5$. The n_f and n_z indicate the number of masked functional and notional words. The notional and functional words are manually labeled for these two examples. The "□" in green color indicates a masked notional word and the "□" in red color indicates a masked functional word, respectively. The word with a yellow shadow is replaced with a random word. The number rate between functional words and notional words and the confidence of the discriminator are given. The sample with the lowest confidence is selected as the final sample.

6.1. Training GAN-MM

We trained the GAN-MM on the Buddhist dataset until the validation loss reached a stable value. We assessed the network's performance with varying settings for N, specifically $N \in 3, 4, 5, 6$. The optimization of the generator and discriminator losses was conducted using the Adam optimizer, employing a learning rate of 1e−3 and a weight decay of 0.01.

The validation loss curves are depicted in Fig. 2. We can observe that the discriminator reaches convergence in fewer than 10 epochs, and the convergence trends across different N settings show no significant differences. In terms of the reconstruction loss and the generator loss, all models converge within 9 and 15 epochs, respectively. Notably, the network training exhibits faster convergence as N increases, until $N \geq 5$, indicating that a deeper network structure enhances the training process. Consequently, we have selected the model trained with $N = 5$ for the purpose of generating samples for pre-training the BERT model in the rest experiments, following the procedure outlined in Algorithm 1.

We provide two examples demonstrating the utilization of the trained GAN-MM to select the ultimate masked samples, as illustrated in Fig. 3. For each sample, 5 candidate samples are randomly subjected to masking. For each of these candidate samples, we employ the trained generator to recover the word embeddings and utilize the trained discriminator to assign a confidence score. The sample with the lowest confidence score is chosen as the final sample. Observing these two examples, we can discern that the generated confidence scores are directly related to the ratio between functional words and notional words. This observation confirms that the GAN-MM frequently selects samples with a higher proportion of notional words in the masking process.

6.2. Pre-training BERT

In this paper, we utilize the pre-trained models BERT-ancient (Wang & Ren, 2022) and BERT-wwm (Cui et al., 2021). Rather than training these models from scratch, we extend their pre-training by incorporating the Buddhist dataset, leveraging the released version from Hugging Face. The optimization of the loss is carried out using the Adam optimizer, with a learning rate set to 1e−4 and a weight decay of 0.01.

To assess the effectiveness of the GAN-MM, we initially compared the performance of pre-trained BERT models when selecting masked samples from varying numbers of candidate samples, ranging from

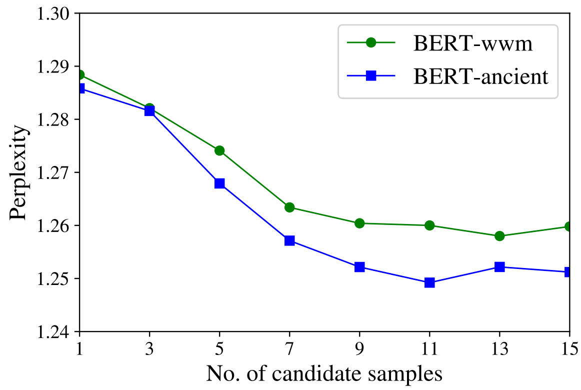


Fig. 4. The Perplexities on the Buddhist validation set.

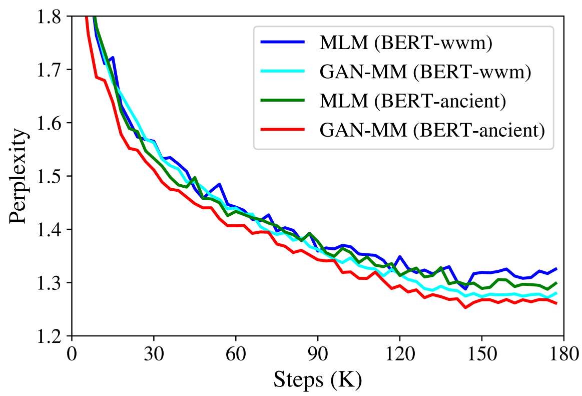


Fig. 5. The Perplexities between MLM and GAN-MM on the Buddhist validation set.

1 to 15, following Algorithm 1. It is crucial to note that the GAN-MM essentially reverts to the Masked Language Model (MLM) when $m = 1$. We evaluated the models using perplexities as the metric, and we selected the best perplexities from the validation set at the 180K steps, as illustrated in Fig. 4. Notably, as the number of candidate masked samples increased, the BERT models exhibited lower perplexity. For the subsequent experiments, we chose to use 11 candidate samples, denoted as $m = 11$. We proceeded to compare the perplexity performance between the MLM and GAN-MM methods during the pre-training of BERT-ancient and BERT-wwm models. It is important to emphasize that these models were not trained from scratch but rather continued training by loading parameters from Hugging Face. The validation perplexity curves are depicted in Fig. 5. The results clearly indicate that the GAN-MM outperforms the baseline MLM method in terms of perplexity during the pre-training of both BERT-ancient and BERT-wwm models. This finding validates the beneficial impact of the proposed GAN-MM in the pre-training of BERT models. Furthermore, as the backbone BERT model, BERT-ancient consistently outperforms BERT-wwm.

6.3. Experimental results on the ZSC dataset

The objective of the ZSC dataset is to distinguish between positive and negative Zen texts. Due to the dataset's limited size, we froze the parameters of all BERT backbones and fine-tuned only the last linear classification layers. Initially, the learning rate was set to 1e−3 and gradually decayed to 1e−4 at the 10th epoch and 1e−5 at the 20th epoch. Learning was terminated after 30 epochs. We optimized the binary cross-entropy loss using Adam with a weight decay of 0.01. Performance evaluation was based on accuracy and F1 score metrics.

Table 2
The accuracy (%) and F1 score (%) on the ZSC dataset.

Models	Pre-training	Accuracy	F1
BERT-base (Devlin et al., 2018)	✗	78.44	76.83
AnchiBERT (Tian et al., 2021)	✗	79.45	76.84
MacBERT-base (Cui et al., 2020)	✗	80.26	78.92
SIKU-BERT (Wang et al., 2022)	✗	80.12	78.37
BERT-wwm (Cui et al., 2021)	✗	80.37	79.31
	MLM	81.02	79.87
	GAN-MM (ours)	**81.68**	**80.24**
BERT-ancient (Wang, 2022)	✗	80.54	78.49
	MLM	81.24	80.01
	GAN-MM (ours)	**82.39**	**81.68**

Table 3
The Recall (%) and F1 score (%) on the ZTS dataset. "✗" indicates that the model is not pre-trained on the Buddhist dataset.

Models	Pre-training	Recall	F1
BERT-base (Devlin et al., 2018)	✗	86.84	93.02
AnchiBERT (Tian et al., 2021)	✗	87.14	93.18
MacBERT-base (Cui et al., 2020)	✗	89.07	94.22
SIKU-BERT (Wang et al., 2022)	✗	89.12	94.23
BERT-wwm (Cui et al., 2021)	✗	89.19	94.28
	MLM	90.24	94.87
	GAN-MM (ours)	**90.65**	**95.32**
BERT-ancient (Wang, 2022)	✗	89.41	94.41
	MLM	90.36	94.99
	GAN-MM (ours)	**90.98**	**95.81**

The experimental results are presented in Table 2. Without pre-training on the Buddhist dataset, BERT-wwm and BERT-ancient outperform other baseline models. When pre-training BERT-wwm, the general MLM increases accuracy by 0.65% and F1 by 0.56%, while the proposed GAN-MM increases accuracy by 1.31% and F1 by 0.93%, respectively. In the case of pre-training BERT-ancient, the general MLM improves accuracy by 0.70% and F1 by 1.52%, and the proposed GAN-MM enhances accuracy by 1.85% and F1 by 3.19%, respectively. Overall, BERT models pre-trained with GAN-MM on the Buddhist dataset achieve the best performance in Zen text classification.

6.4. Experimental results on the ZTS dataset

The objective of the ZTS dataset is to segment Zen texts. In each sample, punctuation marks are omitted, and the labels consist of a sequence of "0" and "1", where "1" indicates the end of a sentence token. The performance metrics utilized are recall and F1 score. A true positive (TP) sample is obtained when a target "1" is predicted, while a false negative (FN) sample is obtained to the contrary. Similarly, a true negative (TN) sample is acquired when a target "0" is predicted, with a false positive (FP) sample obtained in the opposite case. The learning rate is initially set to 1e−3 and subsequently decayed to 1e−4 at the 10th and 30th epochs. Learning is terminated after 50 epochs. We optimize the binary cross-entropy loss using Adam with a weight decay of 0.01.

The experimental results are presented in Table 3. Without pre-training on the Buddhist dataset, BERT-wwm and BERT-ancient outperform other baseline models. When pre-training BERT-wwm, the general MLM increases recall by 1.05% and F1 by 0.59%, while the proposed GAN-MM increases recall by 1.46% and F1 by 1.04%, respectively. For pre-training BERT-ancient, the general MLM improves recall by 0.95% and F1 by 0.58%, and the proposed GAN-MM enhances recall by 1.57% and F1 by 1.40%, respectively. Overall, BERT models pre-trained with GAN-MM on the Buddhist dataset achieve the best Zen text segmentation performance.

Table 4
The accuracy (%) on the PTC dataset.

Models	Pre-training	Accuracy
Std-Transformer (Vaswani et al., 2017)	✗	69.96[a]
BERT-base (Devlin et al., 2018)	✗	75.31[a]
AnchiBERT (Tian et al., 2021)	✗	82.30[a]
AnchiBERT (Tian et al., 2021)	MLM	82.32
	GAN-MM (ours)	**82.58**
BERT-wwm (Cui et al., 2021)	MLM	80.24
	GAN-MM (ours)	80.34
BERT-ancient (Wang, 2022)	MLM	79.49
	GAN-MM (ours)	79.85

[a] Indicates that the results are from Tian et al. (2021).

Table 5
The BLEU-1 and BLEU-2 on the CCG dataset. "–" indicates that the corresponding results are not given.

Models	Pre-training	BLEU-1	BLEU-2
S2S (Yuan, Zhong, Li, & Zhang, 2019)	✗	19.49[b]	–
S2SA (Yuan et al., 2019)	✗	19.68[b]	–
CoupGAN (Qu, Lv, Liu, & Yang, 2022)	✗	22.43[b]	–
NCM (Yan, Li, Hu, & Zhang, 2016)	✗	20.82[b]	20.55[a]
Std-Transformer (Vaswani et al., 2017)	✗	–	27.14[a]
BERT-base (Devlin et al., 2018)	✗	–	33.01[a]
AnchiBERT (Tian et al., 2021)	✗	–	33.37[a]
AnchiBERT (Tian et al., 2021)	MLM	34.43	33.33
	GAN-MM (ours)	**35.12**	**33.55**
BERT-wwm (Cui et al., 2021)	MLM	23.45	22.40
	GAN-MM (ours)	23.82	23.28
BERT-ancient (Wang, 2022)	MLM	23.44	22.79
	GAN-MM (ours)	23.64	23.38

[a] Indicates that the results are from Tian et al. (2021).
[b] Indicates that the results are from Qu et al. (2022).

6.5. Experimental results on the PTC dataset

To evaluate the proposed method for ancient Chinese poem topic classification tasks, we conducted a comparison with state-of-the-art (SOTA) methods, including the standard Transformer (Std-Transformer), which represents the official BERT-base without pre-training, BERT-base, and AnchiBERT. AnchiBERT, BERT-wwm, and BERT-ancient were pre-trained using MLM and GAN-MM on the Buddhist dataset, respectively. In this task, the input consists of a sequence of poems, and the output is the corresponding topic label. We optimized the cross-entropy loss using the Adam optimizer with a learning rate of 5e−5, as referenced in Tian et al. (2021). The weight decay is set to 1e−2, and the fine-tuning process typically spans around 10 epochs.

The experimental results are presented in Table 4. GAN-MM demonstrated accuracy improvements of 0.26%, 0.10%, and 0.36% when used for pre-training Ancient-BERT, BERT-wwm, and BERT-ancient, respectively, compared to MLM. The top-performing SOTA method is AnchiBERT, achieving an accuracy of 82.58%. In summary, pre-training on the Buddhist dataset can achieve comparable accuracy performance to SOTA methods.

6.6. Experimental results on the CCG dataset

To evaluate the proposed method for ancient Chinese couplet generation tasks, we conducted a comparison with SOTA methods, including S2S (Yuan et al., 2019), S2SA (Yuan et al., 2019), CoupGAN (Qu et al., 2022), NCM (Yan et al., 2016), Std-Transformer, BERT-base, and AnchiBERT. The evaluation metrics used were BLEU-1 and BLEU-2 (Papineni, Roukos, Ward, & Zhu, 2002). Following the experimental setup described in AnchiBERT (Tian et al., 2021), we employed a Transformer-based decoder, initialized randomly, with a layer number set to 4. The optimization was carried out using the Adam optimizer, and a linear warm-up of over 4000 steps was applied. The initial learning rate was set to 1e−3 and then decayed to 1e−4 at the 30th epoch. The fine-tuning process concluded after 60 epochs.

The experimental results are given in Table 5. Concerning BLEU-1, AnchiBERT, BERT-wwm, and BERT-ancient, which are pre-trained on the Buddhist dataset, outperform SOTA models. However, when it comes to BLEU-2, BERT-wwm and BERT-ancient do not achieve results comparable to those of Std-Transformer, BERT-base, and AnchiBERT, as reported in Tian et al. (2021). This disparity can be attributed to the fact that these models were pre-trained on an ancient Chinese dataset consisting of poems and couplets, whereas the Buddhist dataset is less relevant to couplet data. Nevertheless, the proposed GAN-MM pre-training method yields better results for both BLEU-1 and BLEU-2 compared to the general MLM. Notably, AnchiBERT with GAN-MM pre-training achieves the highest BLEU-1 of 35.12% and BLEU-2 of 33.55%.

7. Conclusion

In this paper, we introduce a novel masking model, named GAN-MM, designed for pre-training BERT models. We have also constructed and publicly released a large-scale Buddhist benchmark dataset for model pre-training. GAN-MM is trained using an innovative semi-supervised learning algorithm. Our experiments demonstrate that GAN-MM outperforms the general MLM on two private Buddhist literature understanding tasks and surpasses other SOTA methods.

Furthermore, we assess the generalization ability of BERT models, which were pre-trained on the Buddhist dataset, on two publicly available ancient Chinese understanding tasks, and the experimental results indicate their comparable performance. Despite the time spent in the GAN-MM training process for selecting final masked samples, it offers two noteworthy advantages for pre-training BERT models. Firstly, GAN-MM achieves rapid convergence in approximately 10 epochs on the Buddhist dataset. Secondly, GAN-MM performs effectively with only 5 layers of transformer blocks. In summary, this paper introduces a significant masking model for pre-training BERT models, achieved through GAN-MM with a semi-supervised learning algorithm.

8. Discussion

This paper has constructed and published a benchmark dataset for Buddhism, along with two ancient Chinese language understanding tasks. Researchers in the field of Buddhist studies can utilize this dataset to perform comprehensive analyses of Buddhist texts, doctrines, and practices. They can delve into historical trends, linguistic variations, and the evolution of Buddhist thought. Linguists and computational linguists may employ natural language processing techniques to scrutinize the language employed in Buddhist scriptures. This could provide valuable insights into the translation of Buddhist texts and the variations found across different languages and dialects.

While the ChatGPT large-scale model (Radford, Narasimhan, Salimans, Sutskever, et al., 2018) has made waves across the globe, comprehending ancient Chinese remains a formidable challenge. Therefore, our efforts aim to contribute to the development of a large-scale ancient Chinese model. We introduce a GAN-MM approach for pre-training BERT models on the Buddhist dataset, initiating a research paradigm for addressing ancient Chinese comprehension. We anticipate that this work will serve as an inspiration for other endeavors related to antiquity.

We conclude that the proposed GAN-MM has two potential limitations. First, GAN-MM is a two-stage learning method, requiring separate training before pre-training the language model. This two-stage approach is time-consuming in both application and deployment. Our future work will focus on developing an end-to-end training method that integrates GAN-MM with the language model. Second, although

GAN-MM is an unsupervised learning method, it relies on manually provided prior knowledge of functional and notional words. This requirement makes it challenging to adapt the GAN-MM training method to other literary corpora, as it necessitates a specialized understanding of the language.

CRediT authorship contribution statement

Chaowen Yan: Investigation, Data curation, Formal analysis, Writing – original draft. **Yong Wang:** Conceptualization, Project administration, Supervision. **Lili Chang:** Investigation, Data curation. **Qiang Zhang:** Conceptualization, Investigation. **Tao He:** Conceptualization, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset is available at https://data.mendeley.com/datasets/5hzs8w46jh/1.

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